

PE-EC701A — Microwave Theory and Technique

Semester VII • 3L:0T:0P • 3 Credits

Module 8 Exam Notes: Microwave Measurements

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1. Power Measurement

Microwave power is categorized based on intensity levels: Low Power ($< 10 \text{ mW}$), Medium Power ($10 \text{ mW} - 10 \text{ W}$), and High Power ($> 10 \text{ W}$).

1.1 Bolometer Mount Method (Low Power)

A bolometer is a temperature-sensitive resistor whose resistance changes when it absorbs RF power.

- **Thermistor:** Has a **negative temperature coefficient (NTC)**; resistance decreases as temperature rises.
- **Barretter:** Has a **positive temperature coefficient (PTC)**; resistance increases as temperature rises.

The bolometer is placed in an active arm of a self-balancing Wheatstone bridge. The DC bias power required to balance the bridge is measured before and after applying RF power:

$$P_{\text{RF}} = P_{\text{bias},1} - P_{\text{bias},2} \quad (1)$$

1.2 Calorimetric Method (High Power)

High power is measured by absorbing the RF energy in a fluid load (usually water) and measuring the temperature rise ($\Delta T = T_{\text{out}} - T_{\text{in}}$).

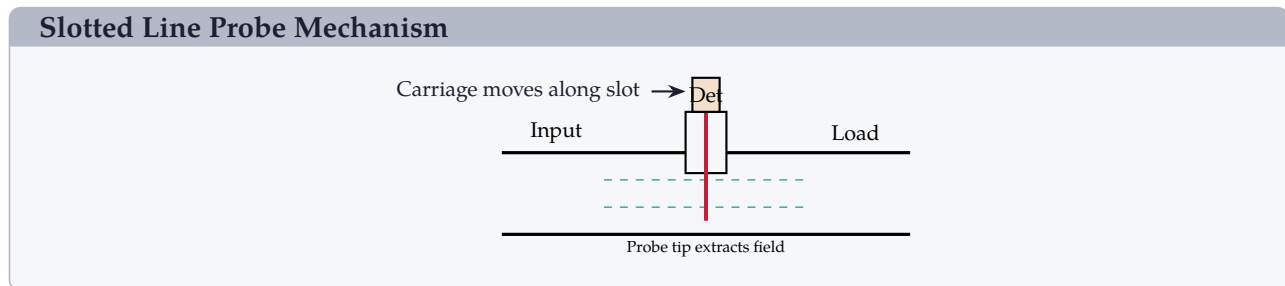
$$P_{\text{RF}} = \dot{m} C_p \Delta T \quad (2)$$

where $\dot{m}$ is mass flow rate and C_p is specific heat.

2. Impedance and Frequency Measurement

2.1 Slotted Line Probe Technique

A slotted line is a section of coaxial line or waveguide containing an axial slot that allows a small capacitive probe to slide along the length to sample the electric field.



- **Low VSWR Measurement** ($1 < S < 10$): Measured directly as the ratio of maximum to minimum probe outputs:

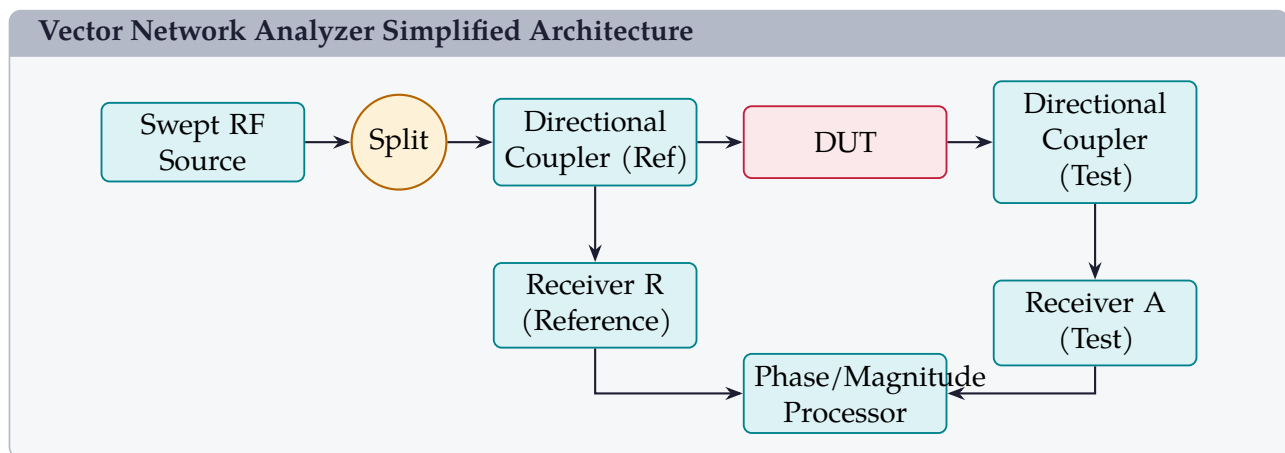
$$S = \frac{V_{\text{max}}}{V_{\text{min}}} \quad (3)$$

- **High VSWR Measurement (Double-Minimum Method, $S > 10$):** For high reflection, the minimum is sharp and close to the noise floor. We measure the distance Δx between two positions where output power is twice the minimum value:

$$S = \frac{\lambda_g}{\pi \Delta x} \quad (4)$$

3. Vector Network Analyzer (VNA)

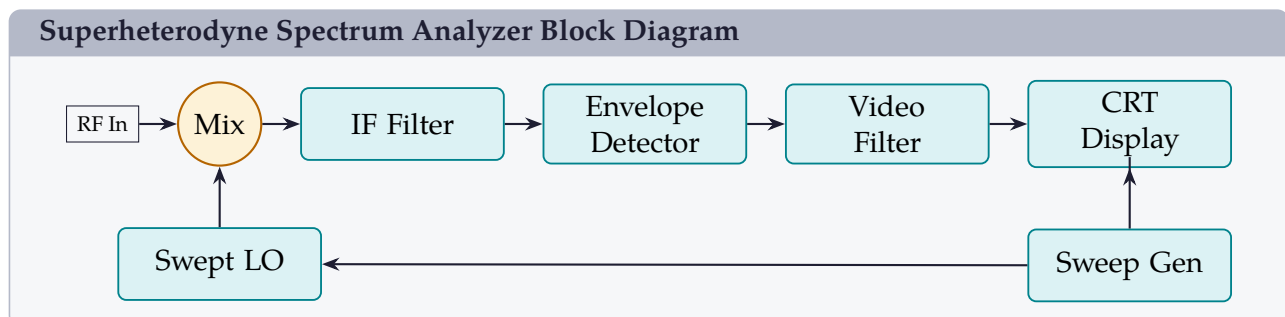
A Vector Network Analyzer measures the complete S-parameters (both magnitude and phase) of a device under test (DUT) by separating incident, reflected, and transmitted waves.



Calibration: System errors are calibrated out using the **SOLT** (Short, Open, Load, Thru) standard calibration procedure.

4. Spectrum Analyzer

A Spectrum Analyzer displays signal amplitude versus frequency, identifying harmonics, distortion, and noise floors.



5. Noise Measurement: Y-Factor Method

The Y-factor method is the standard technique for measuring the noise figure (F) of a receiver:

Y-Factor Method Equations

We measure output noise power with a hot noise source at temperature T_h and a cold noise source at temperature T_c . The ratio is:

$$Y = \frac{N_{\text{hot}}}{N_{\text{cold}}} = \frac{T_h + T_e}{T_c + T_e} \quad (5)$$

Solving for the equivalent noise temperature T_e of the receiver:

$$T_e = \frac{T_h - YT_c}{Y - 1} \quad (\text{K}) \quad (6)$$

The noise figure F is then calculated as:

$$F = 1 + \frac{T_e}{T_0} \quad (\text{with } T_0 = 290 \text{ K}) \quad (7)$$

6. Measurement of Microwave Antenna Parameters

Antenna parameter measurements are critical for validating design parameters such as radiation pattern and directive gain.

6.1 Radiation Pattern Measurement

The radiation pattern is measured in the far-field region to ensure that the wave behaves as a local plane wave.

- **Far-Field Boundary (Fraunhofer Distance):** The minimum distance R between the transmitting antenna and the antenna under test (AUT) of maximum dimension D is:

$$R \geq \frac{2D^2}{\lambda} \quad (\text{m}) \quad (8)$$

- **Setup:** Carried out in an **anechoic chamber** lined with RF-absorbing foam pyramids to eliminate echoes. The AUT is placed on a rotating positioner as a receiver, and a stationary source antenna transmits a signal. The received power is recorded as a function of the rotation angle (θ, ϕ) to plot the 2D/3D radiation pattern.

6.2 Antenna Gain Measurement

Gain is measured using either absolute or relative methods:

1. **Three-Antenna Method (Absolute Method):** Requires no prior knowledge of any antenna's gain. By measuring the received power combinations (P_{ij}) among three different antennas (A, B, C) separated by distance R , we apply the **Friis Transmission**

Equation to solve for individual gains:

$$G_i G_j = \left(\frac{4\pi R}{\lambda} \right)^2 \frac{P_{rec,j}}{P_{trans,i}} \quad (9)$$

Taking the logarithm yields a system of three linear equations to find G_A, G_B, G_C .

2. **Gain Transfer Method (Comparison Method):** The received power of a standard gain horn antenna (with known gain G_{std}) is measured first as P_{std} . The AUT is then substituted in the exact same position, and its received power P_{AUT} is measured. The gain of the AUT is calculated as:

$$G_{AUT} = G_{std} \cdot \frac{P_{AUT}}{P_{std}} \quad \Rightarrow \quad G_{AUT(dBi)} = G_{std(dBi)} + 10 \log_{10} \left(\frac{P_{AUT}}{P_{std}} \right) \quad (10)$$

Quick Revision — Module VIII

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Key Formulas at a Glance

Low VSWR	$S = \frac{V_{\max}}{V_{\min}}$
Double-Minimum VSWR	$S = \frac{\lambda_g}{\pi \Delta x} \quad (\Delta x \text{ is distance between } 2V_{\min} \text{ points})$
Calorimetric Power	$P = \dot{m}C_p(T_{\text{out}} - T_{\text{in}})$
Y-Factor Relation	$Y = \frac{N_{\text{hot}}}{N_{\text{cold}}} = \frac{T_h + T_e}{T_c + T_e}$
Receiver Noise Temp	$T_e = \frac{T_h - YT_c}{Y - 1}$

One-Line Definitions

- **Thermistor:** A temperature-sensitive resistor with a negative temperature coefficient (NTC).
- **Barretter:** A thin platinum wire bolometer element with a positive temperature coefficient (PTC).
- **Slotted Line:** A waveguide with an axial slot used to sample electric field standing wave patterns.
- **VNA:** An instrument that measures the amplitude and phase of reflection/transmission S-parameters.
- **Spectrum Analyzer:** A swept receiver displaying power spectral density versus frequency.
- **Y-Factor:** The ratio of receiver output noise powers under hot and cold source excitations.
- **SOLT Calibration:** Standard VNA calibration using Short, Open, Load, and Thru reference elements.

Frequently Asked / Viva Questions

Q1. Explain why a bolometer Wheatstone bridge must be self-balancing.

Applying RF power heats the bolometer, changing its resistance and unbalancing the bridge. Automatically reducing DC bias to re-balance the bridge allows RF power to be measured from the DC power difference, keeping bolometer resistance constant.

Q2. What is the double-minimum method and when is it used?

It is a method to measure high VSWR ($S > 10$). We measure the distance Δx between the 3 dB points around a standing-wave minimum. It is used because direct $V_{\max}/V_{\min}$ measurements are inaccurate near the noise floor.

Frequently Asked / Viva Questions (contd.)**Q3. What is the difference between a scalar and a vector network analyzer?**

A Scalar Network Analyzer (SNA) only measures amplitude transmission and reflection coefficients. A Vector Network Analyzer (VNA) measures both amplitude and phase.

Q4. Why does a frequency wavemeter have a resonant cavity?

It is a tunable cavity resonator. When tuned to the input frequency, it absorbs a fraction of power, creating a dip in the through-line power meter to indicate frequency.

Q5. Explain the SOLT calibration method.

VNA calibration standards: Short circuit (reflects phase 180°), Open circuit (reflects phase 0°), matched Load (absorbs all power), and Thru connection (connects ports directly).

Q6. Describe the basic block diagram of a superheterodyne spectrum analyzer.

RF input is mixed with a swept LO. The resulting IF signal is filtered, envelope-detected, and displayed on a screen swept in sync with the LO.

Q7. How is antenna gain measured using the comparison method?

The received power of a standard gain horn (known gain G_s) is measured. The test antenna is then substituted, and its received power P_t is measured. The test gain is $G_t = G_s(P_t/P_s)$.

Q8. Why is calorimetric power measurement used for high powers?

Because solid-state sensors (bolometers/diodes) saturate or burn out at high powers. Fluid loads can absorb kilowatts of power safely, converting it to measurable heat.

Q9. What is the physical meaning of the equivalent noise temperature (T_e)?

It is the temperature of a dummy resistor at the input of an ideal noiseless receiver that would generate the same noise power as the actual noisy receiver.

Q10. Why is the slot in a slotted line waveguide cut along the center of the broad wall?

Because the slot is parallel to the surface current flow lines of the dominant TE_{10} mode. This minimizes field distortion and leakage.

Q11. What are the typical noise sources used in Y-factor measurements?

Gas discharge tubes or avalanche diodes that can switch between a room temperature state (cold) and a high electron temperature state (hot).

Q12. How does a Spectrum Analyzer differ from an Oscilloscope?

An oscilloscope displays signal amplitude versus time (time domain). A spectrum analyzer displays signal amplitude versus frequency (frequency domain).

Comparison of VNA and Spectrum Analyzer

Feature	Vector Network Analyzer (VNA)	Spectrum Analyzer
Signal Source	Built-in swept source (active stimulus).	None (passive measurement of external signal).
Quantities Measured	S-parameters (S_{11} , S_{21} , etc.), phase, group delay.	Frequency component amplitudes, harmonics, noise floor.
Calibration	Extremely complex (SOLT calibration required).	Simple amplitude calibration.
Primary Use	Characterizing passive/active components (filters, antennas).	Analyzing transmitter signals, interference, distortion.